Estimating and Hedging Exchange Rate Exposure

Automatic Data Processing, Inc. (ADP) is a global leader in human capital management (HCM) solutions, providing payroll, HR, talent, time, tax, and benefits administration to businesses of all sizes. Founded in 1949 and headquartered in the U.S., ADP serves over 1 million clients in more than 140 countries on innovative cloud-based technology. With decades of experience, a commitment to data security, and a focus on delivering actionable insights, ADP continues to shape the future of work for its diverse global client base (ADP 2023). As investors, it's critical to understand its business landscape and exchange risk exposure.

1. Currency Beta Analysis

To estimate ADP's systematic risk level to the market and the USD, a multi-factor regression model is done based on this equation (*):

$$R_{i,t} - R_{f,t} = \alpha + \beta_{Mkt} \times (R_{Mkt,t} - R_{f,t}) + \beta_{FX} \times R_{FX,t} + \epsilon_{i,t} \ (*)$$

The model generated the estimated $\beta_{Mkt} = 0.941$, β_{FX}, = 0.429 and an adjusted R-squares of 42.4%.

The coefficient of the DXY return β_{FX} is positive and statistically significant (p-value < 0.05), implying that for every 1% increase in the USD index, ADP's excess return rises by approximately 0.429%. This positive currency beta suggests that ADP's performance tends to move in the same direction as USD appreciates, which may be beneficial for the firm's revenue from the U.S. market, or investor sentiment towards its stock listed in the U.S. market. However, β_{FX} is notably smaller than that of the market, suggesting a moderate sensitivity to currency fluctuations compared with broader market movements.

Similarly, the market excess return coefficient β_{Mkt} is more than double that of the currency beta, and is also statistically significant (p-value <0.05), confirming that overall equity market movements exert a stronger influence on ADP's returns than fluctuations in USD exchange rates. This aligns with Capital Asset Pricing Model (CAPM) theory, where market risk is the primary systematic driver of stock returns, while currency beta acts as an additional, secondary risk factor. Nonetheless, β_{FX} remains relevant, especially for firms with substantial foreign exchange exposure, as currency swings can erode or enhance overseas revenues when translated into USD.

Overall, the model explains 42.4% of the variation in ADP's excess returns, which is a reasonable explanatory power given that equity prices are also influenced by idiosyncratic factors such as firm-specific news, sector trends, and short-term market shocks. The statistical significance of both betas and the model reinforces the importance of considering not only traditional market risk but also exchange rate movements to address ADP's evolving and dynamic risk landscape.

2. Business and Operational Effects on Currency Beta

ADP-implied currency beta is positive yet poses a moderate level of systematic risk to the company's operations.

First of all, ADP's operations are predominantly U.S.-based, with 88.56% of total revenues generated in USD. Europe is its second-largest market, though it has exhibited a steady, dampened revenue trend in recent years, while Canada and other regions show only marginal growth (ADP 2023). This geographic distribution implies that, although ADP is a MNC with an established presence across multiple countries, its earnings base is mostly USD-denominated. As a result, its overall exposure to FX risk is moderate and of little concerns.

Moreover, the net assets of ADP's foreign subsidiaries are translated into U.S. dollars at period-end exchange rates, while revenues and expenses are translated at average exchange rates over the reporting period (ADP 2023). This translation method, combined with the small proportion of foreign-currency-denominated operations, reduces the potential for significant earnings volatility, leading to minimal FX exposure and low beta.

The company's 2023 financials also highlight a notable change in ADP's currency translation adjustment despite stable revenue stream derived from foreign operations. This gap could potentially be attributed to external macroeconomic factors such as the pronounced weakening of certain subsidiary currencies like EUR against USD during the year (Bibey 2023, Export Finance Australia 2023). However, given that most ADP's revenue is denominated in USD, the impact of such currency depreciation was ultimately minimal.

In terms of its risk management, ADP follows a disciplined approach by utilizing a combination of operational and financing strategies, supplemented by derivative financial instruments when necessary (ADP 2023). Importantly, derivatives are arranged solely for hedging purposes and not for speculative gains, aligning with a conservative treasury policy that aims to stabilize cash flows and earnings against exchange rate volatility. This approach effectively stabilizes cashflows and limits the company's sensitivity to FX volatility, and thus remain a low beta.

Overall, ADP's positive but moderate currency beta reflects its partial exposure to foreign exchange fluctuations while remaining protected from excessive volatility due to its USD-centric revenues and active risk management strategies.

3. ADP Stock Performance Prediction using CAPM

Using equation (*) and 2024 cumulative market excess returns, DXY returns and risk-free rate, the predicted ADP returns is 0.227, which is -3.61% different than the actual 2014 returns. This prediction error is generally acceptable, indicating a fairly robust model to forecast future ADP returns.

This overestimation firstly is due to a slightly underperformed market in 2024, compared to the previous year (Pan 2023). With a positive market beta as estimated previously, this led to a lower market factor. In contrast, USD was doing better in 2024 compared to the previous year and the sample period (State Street Investment Management 2024). Similarly, with a positive FX beta, this resulted in a strengthened currency factor. As USD appreciation is more pronounced than S&P500 underperformance, this led to the overestimation in stock returns.

Nevertheless, it's critical to acknowledge the limitation of CAPM model such as linear relationship, perfectly efficient market and omitted variables related to operational and compliance risks, which is highlighted in the annual report (ADP 2023).

4. Exchange Risk Analysis

To analyze exchange risk, Europe and Canada, which are ADP's largest markets after the US, are chosen as comparison groups. Two regression models (EUR/USD and CAD/USD) are conducted using five independent variables: US GDP growth, counterpart GDP growth, interest rate differential, inflation differential, and account balance differential.

Models	Coefficient estimates (p-value)					Adjusted R-squares	Significance F
	GDP Growth – Canada/ Europe	GDP Growth - US	Interest Rate Differential	Inflation Rate Differential	Account Balance Differential		
CAD/USD	0.105 (0.771)	0.244 (0.544)	0.011 (0.974)	-0.455 (0.013)	-0.118 (0.018)	0.032	0.018
EUR/USD	-0.242 (0.370)	0.581 (0.088)	-0.102 (0.530)	-0.369 (0.021)	-0.208 (0.086)	0.025	0.038

Figure 1: Exchange rate Regression Model Results

Overall, these macroeconomic factors explain only a small portion of exchange rate variations at 2.5% and 3.2%, as shown in Figure 1. However, both models are statistically significant at the 5% level, indicating they still provide meaningful insights into exchange risk. This is reasonable, as macroeconomic variables typically influence exchange rates over the long term, so they are less capable of capturing short-term fluctuations or shifts driven by market sentiment.

From the regression results, it's apparent that the two main drivers that are mostly statistically significant, are account balance differential and inflation rate differential.

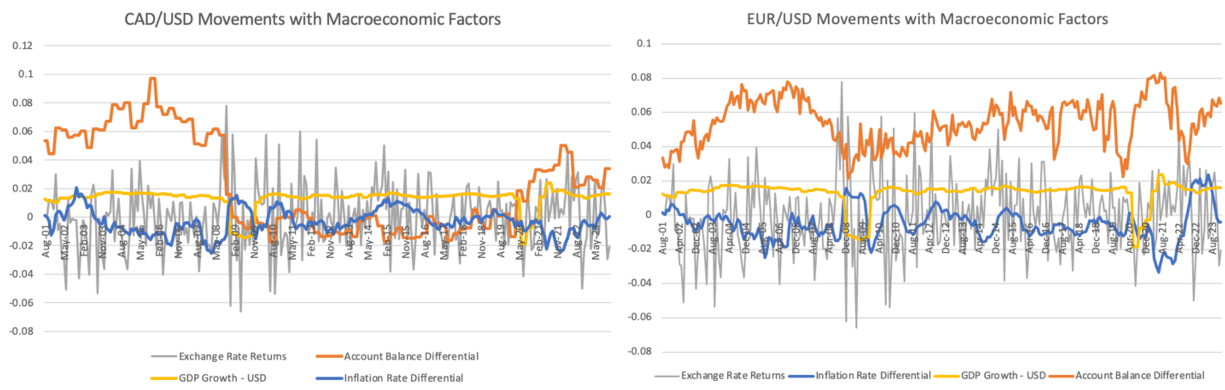

Figure 2: Exchange Rate movements with Key Macroeconomic Factors

For account balance differential, both models' coefficients are negative, indicating USD appreciation when the U.S. has a larger account surplus compared to Canada or the Eurozone, the USD tends to appreciate, which aligns with the Balance of Payments theory. This relationship is also reflected in Figure 2, particularly in the post-COVID period, where lows in account balance differentials coincided with peaks in USD appreciation. As a result, trade and capital flow are one of the main factors that shape DXY movements.

Inflation rate differential's coefficients are also negative but show the opposite effect. Surprisingly, the USD appreciates when the U.S. inflation rate is higher than that of other countries, which contradicts the Purchasing Power Parity (PPP) theory. As USD is the dominant currency in global trade in one of the most advanced economies, it has a relatively low exchange rate pass-through (ERPT), which isolates USD from inflation pressures (Gratton 2024). From Figure 2, the inflation differential's effect also appears to be lagged and less pronounced, further supporting the view that ERPT reduces the immediate impact of inflation on currency movements. This highlights the complex effect of inflation on DXY movements in modern financial markets.

The last key driver of DXY is GDP growth of the U.S. Despite not statistically significant at the 5% level, it remains economically meaningful compared to other variables. U.S. GDP growth has a positive impact on the exchange rate, indicating USD appreciation during periods of strong economic activity and favorable investment conditions. This positive relationship is observed during major events of the sample period, such as the Global Financial Crisis and the COVID-19 pandemic. Despite weak correlation and relatively indifferent behavior, U.S. GDP growth remains a relevant factor for analyzing DXY movements in key periods.

In general, despite having some macroeconomic key drivers, they still have limited ability to explain DXY movement, highlighting the dynamics and complexity in exchange risk, which could be enhanced with more short-term macroeconomic fundamentals and modern factors.

5. Hedging Strategy

A USD 1 million investment in ADP from July 1 to December 15, 2025, is hedged using ICE U.S. Dollar Index (DX) futures. Given the hedge period, the DEC25 contract at an index price of $97.18 is selected (ICE n.d.). As each DX contract has a notional value of USD 1,000 per index value, the hedge ratio is calculated using the equation below:

$$h = \frac{R_H - R}{R_F}$$

The hedge ratio is estimated at 4.42 contracts. The stock's positive currency beta indicates that a depreciation of the USD would negatively impact returns, so the appropriate hedge is a long position in USD Index futures. For practical implementation, hedging against this investment requires buying 5 ICE U.S. Dollar Index (DX) DEC25 futures contracts.

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